§9. 1000题&习题总结.

1000.9.1(1). 见根号 → 凑平方.

$$\int \frac{1}{\sqrt{x(4-x)}}dx \overset{令x=4\sin^2 t}{=\!=\!=} \int \frac{4\cdot 2\sin t\cdot\cos t}{2\sin t\cdot 2\cos t}dt$$

$$= \int 2dt = \cancel{2x}\ 2t+C.\quad \sin t = \sqrt{\frac{x}{4}}\quad t=\arcsin\sqrt{\frac{x}{4}}\ \big|x.$$

$$\int \frac{1}{\sqrt{x(4-x)}}dx \qquad \sqrt{x(4-x)} = \sqrt{-(x^2-4x)} = \sqrt{4-(x-2)^2} = 2\sqrt{1-\left(\frac{x-2}{2}\right)^2}$$

$$= \int \frac{dx}{2\sqrt{1-\left(\frac{x-2}{2}\right)^2}}$$

$$= \arcsin\frac{x-2}{2} + C$$

1000.9.1(2)

$$\int \sin^3 x\cos^2 x\,dx$$

$$= -\int \cos^2 x\sin^2 x\,d\cos x$$

$$= -\int \cos^2 x(1-\cos^2 x)\,d\cos x = \int\cos^4 x\,d\cos x - \int\cos^2 x\,d\cos x$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

1000.9.1(3).

$$\int\frac{2x-6}{x^2-6x+13}dx$$

$$= \int\frac{d(x^2-6x)}{(x^2-6x)+13} = \int\frac{d(t+13)}{t+13} = \ln|x^2-6x+13| + C$$

1000.9.1(4).

$$\int\frac{1}{\sin^2 x\cos^4 x}dx$$

$$= \boxed{\int\frac{(\sin^2 x+\cos^2 x)^2}{\sin^2 x\cdot\cos^4 x}dx}$$

$$= \int \frac{\sin^2 x}{\cos^4 x} + \frac{1}{\sin^2 x} + \frac{2}{\cos^2 x}\,dx$$

$$= \int \sin^2 x\sec^4 x + 2\sec^2 x + \csc^2 x\,dx$$

$$= \int\tan^2 x\,d\tan x + 2\int\sec^2 x\,dx + \int\csc^2 x\,dx$$

$$= \frac{1}{3}\tan^3 x + 2\tan x - \cot x + C$$

1000.9.1(5)

$$\int\frac{\sin x-3\cos x}{\sin^3 x}dx = \int\csc^2 x\,dx - 3\int\frac{d\sin x}{\sin^3 x}$$

$$= -\cot x - \frac{3(\sin x)^{-3}}{-2} = -\cot x + \frac{3}{2}\csc^3 x + C$$

1000.9.1(6).

$$\int\frac{x^3}{\sqrt{1-x^2}}dx. \overset{令x=\cos t}{\underset{t=\arccos x.}{}}\quad \sin t=\sqrt{1-x^2}$$

$$= \int\frac{-\cos^3 t\cdot\sin t}{\sin t}dt = -\int\cos^2 t\,d\sin t$$

$$= \int\sin^2 t\,d\sin t - \int\cos t\,dt = \frac{\sin^3 t}{3} - \sin t + C_1$$

$$= \frac{(1-x^2)^{\frac{3}{2}}}{3} - \sqrt{1-x^2} + C$$

1000.9.2.

设 $f'(\ln x)=\begin{cases}1, & 0<x\le 1\\ x, & x>1\end{cases}$ 且 $f(0)=0$. 则 $f(x)=$?

[solution].

令 $\ln x = t$. $x\in(0,1]$. $t\in(-\infty,0]$

$f'(t)=1 \rightarrow f(t)=t+C$, $t\in(-\infty,0]$

$f(x)=x+C$.

$f(0)=0 \rightarrow C=0$.

$x\in(1,+\infty) \rightarrow t\in(0,+\infty)$.

$f'(t)=e^t \rightarrow f(x)=e^x+C$.

$\lim_{x\to 0^+} f(x)=0 \rightarrow C=-1$

$\therefore f(x)=\begin{cases}x, & x\le 0\\ e^x-1, & x>0\end{cases}$

1000.9.4.

$$\int \frac{x\ln(x+\sqrt{1+x^2})}{(1+x^2)^2}dx$$

$$=-\frac{1}{2}\int \ln(x+\sqrt{1+x^2})\, d\,\frac{1}{1+x^2}$$

$$=-\frac{1}{2}\frac{\ln(x+\sqrt{1+x^2})}{1+x^2}+\frac{1}{2}\int\frac{d\ln(x+\sqrt{1+x^2})}{1+x^2}$$

$$=\frac{-\ln(x+\sqrt{1+x^2})}{2(1+x^2)}+\frac{1}{2}\int\frac{(1+x^2)^{\frac{1}{2}}-x^2(1+x^2)^{-\frac{1}{2}}}{(1+x^2)^1}dx$$

$$=\cancel{\frac{-\ln(x+\sqrt{1+x^2})}{2(1+x^2)}+\frac{1}{2}\int\frac{1}{\sqrt{1+x^2}}dx-\frac{1}{2}\int\frac{x^2dx}{(1+x^2)^{\frac{3}{2}}}}$$

$x=\tan t$: $\cancel{\frac{-\ln(x+\sqrt{1+x^2})}{2(1+x^2)}+\frac{1}{2}\ln(x+\sqrt{x^2+1})-\frac{1}{2}\int\frac{\tan^2t\sec^2t\,dt}{\sec^3t}}$

$$=\frac{-1+(1+x^2)}{2(1+x^2)}\ln(x+\sqrt{x^2+1})-\frac{1}{2}\ln|\sec t|+C$$

$\downarrow \sec t=\sqrt{1+x^2}$

$$=\frac{x^2}{2(1+x^2)}\ln(x+\sqrt{x^2+1})-\frac{1}{2}\ln\sqrt{1+x^2}+C$$

$\sin t=\frac{x}{1+x^2}$

$$=\frac{-\ln(x+\sqrt{1+x^2})}{2(1+x^2)}+\frac{1}{2}\int\frac{dx}{(1+x^2)^{\frac{3}{2}}}$$

$$=\frac{-\ln(x+\sqrt{1+x^2})}{2(1+x^2)}+\frac{1}{2}\sqrt{\frac{x^2}{1+x^2}}+C$$

1000.9.5.

$$\int_0^3 \arcsin\sqrt{\frac{x}{1+x}}dx$$

$$=\cancel{\int_0^{\frac{\sqrt3}{2}}\arcsin t\, d\frac{t^2}{t^2-1}=\int \frac{t^2\arcsin t}{t^2-1}\Big|_0^{\frac{\sqrt3}{2}}-\int_0^{\frac{\sqrt3}{2}}\frac{t^2}{t^2-1}\cdot\frac{dt}{\sqrt{1-t^2}}}$$

令 $\arcsin\sqrt{\frac{x}{1+x}}=t \rightarrow \sqrt{\frac{x}{1+x}}=\sin t$ $\quad \frac{x}{1+x}=\sin^2 t$

$$=\int_0^{\frac{\pi}{3}} \cancel{\sin t}\, t\cdot 2\tan t\cdot\frac{1}{\cos^2 t}$$

$$=2\int_0^{\frac{\pi}{3}} t\,\frac{\sin t}{\cos^3 t}dt=\int_0^{\frac{\pi}{3}} t\, d(\cos^{-2}t)$$

$$=t\cdot\cos^{-2}t\Big|_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{3}}\sec^2 t\,dt=\frac{4}{3}\pi-\sqrt3.$$

1000.9.6

$$\int_1^2 \frac{2x^2+x+1}{(2x-1)(2x^2+x-1)}dx$$

$$= \int_1^2 \frac{2x^2+x+1}{(2x-1)^2(x+1)}dx \qquad \begin{cases} A=5/9 \\ B=4/3 \\ C=2/9 \end{cases}$$

$$= \int_1^2 \frac{A}{2x-1} + \frac{B}{(2x-1)^2} + \frac{C}{x+1}\,dx \qquad \left(\frac{1}{2x-1}\right)$$

$$= \frac{A}{2}\ln|2x-1|\Big|_1^2 + C\ln|x+1|\Big|_1^2 + \frac{-B}{2}\,\cancel{\frac{\ln|2x-1|}{2x}}\Big|_1^2$$

$$= \frac{A}{2}\ln 3 + C\ln 3 - C\ln 2 + \frac{B}{3}$$

$$= \frac{1}{2}\ln 3 - \frac{2}{9}\ln 2 + \frac{4}{9}$$

1000.9.7

反常积分 $\int_0^1 \arctan\sqrt{\frac{x}{1-x}}\,dx$

令 $\arctan\sqrt{\frac{x}{1-x}} = t$ $\qquad \sqrt{\frac{x}{1-x}} = \tan t \quad \tan^2 t = \frac{x}{1-x}$

$$x = \frac{\tan^2 t}{\sec^2 t} = \sin^2 t.$$

$\int_0^1 \arctan\sqrt{\frac{x}{1-x}}\,dx$

$$= \int_0^{\frac{\pi}{2}} t\,d\sin^2 t = -\int_0^{\frac{\pi}{2}} \sin^2 t\,dt + t\sin^2 t\Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} + \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos 2t - 1\,dt = \frac{\pi}{2} + 0 \underbrace{- \frac{\pi}{4}}_{=\frac{\pi}{2}} = \frac{\pi}{4}$$

1000.9.8

$$\int_0^{+\infty} \frac{1-e^{-x}}{\sqrt{x^3}}\,dx \qquad x^{-\frac{3}{2}}$$

$x=t^2$

$$\cancel{= \int_0^{+\infty} \frac{1-e^{-t^2}}{t^3}\,2t\,dt = 2\int_0^{+\infty} \frac{1-e^{-t^2}}{t^2}\,dt}$$

$$\cancel{= -2\frac{1}{t}\Big|_0^{+\infty}}$$

$$= \int_0^{+\infty} -2(1-e^{-x})\,d\,x^{-\frac{1}{2}}$$

$$= -2x^{-\frac{1}{2}}(1-e^{-x})\Big|_0^{+\infty} + 2\int_0^{+\infty} x^{-\frac{1}{2}}\,d(1-e^{-x})$$

$$= 0 + 2\int_0^{+\infty} x^{-\frac{1}{2}}e^{-x}\,dx \overset{x=t^2}{=\!=} 4\boxed{\int_0^{+\infty} e^{-t^2}\,dt} \quad \frac{\sqrt{\pi}}{2}$$

$$= 4\cdot\frac{\sqrt{\pi}}{2} = 2\sqrt{\pi}$$

1000.9.9

$\int_0^1 f(x)\,dx = 1 \qquad f(1)=0.$

$$\int_0^1 x f'(x)\,dx = \int_0^1 x\,df(x) = xf(x)\Big|_0^1 - \int_0^1 f(x)\,dx$$

$$= f(1) - 1 = -1$$

1000.9.10.

设 $\frac{\ln x}{x}$ 是 $f(x)$ 一个原函数. $\int_1^e x f(x)\,dx$

$$= \int_1^e x\,df(x) = xf(x)\Big|_1^e - \int_1^e f(x)\,dx = -1 - \left(\frac{1}{e} - 0\right)$$

$$= -1 - \frac{1}{e}$$

$f(x) = \left(\frac{\ln x}{x}\right)' = \frac{1-\ln x}{x^2}$

$xf(x) = \frac{1-\ln x}{x}$

1000.9.11

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{3e^x\sin^2 x}{1+e^x}dx$$

$$\cancel{=3\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x\, d\, 1+e^x}{1+e^x} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 3\sin^2 x\, d\ln|1+e^x|}$$

$$\cancel{= 3\sin^2 x \ln|1+e^x|\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \ln|1+e^x|\cdot 6\sin x\cdot\cos x\, dx}$$

$$\cancel{= 3\ln\left|\frac{1+e^{\frac{\pi}{2}}}{1+e^{-\frac{\pi}{2}}}\right| -}$$

$$\overset{令t=-x}{=} \int_{+\frac{\pi}{2}}^{-\frac{\pi}{2}} \frac{3e^{-t}\sin^2 t}{1+e^{-t}} d(-t) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{3\sin^2 t}{e^t+1}dt$$

$$= \frac{3}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 t\, dt = \frac{3}{4}\pi$$

?

1000.9.12

$$\frac{d}{dx}\left[\int_0^x \sin(x-t)^2 dt\right]$$

$$= \frac{d}{dx}\cancel{\sin}\left[\int_0^x \sin t^2 dt\right] = \sin x^2$$

1000.9.~~13~~4

$$f(x) = e^{-x^2},\ \int_0^1 f'(x)f''(x)dx = ?$$

$$\int_0^1 f'(x)\, df'(x) = \frac{[f'(x)]^2}{2}\Big|_0^1 - \int_0^1 f(x)\, df'(x)$$

$$\Rightarrow A = \frac{1}{2}(e^{-1}\cdot(-2))^2 = 2e^{-2}$$

$$f'(x) = e^{-x^2}\cdot(-2x)$$

1000.9.13.

$$设f(x)连续,\ \frac{d}{dx}\left[\int_0^x tf(x^2-t^2)dt\right] = \frac{d}{dx}\left[\frac{1}{2}\int_0^x f(x^2-t^2)dt^2\right]$$

$$令x^2-t^2=u^2$$

$$= \frac{d}{d\cancel{(x^2+t^2)}x}\left[\int_x^0 \frac{1}{2}f(u^2)\, d(-u^2)\right]$$

$$= \frac{d}{d\cancel{(x^2+t^2)}x}\int_0^x \frac{f(u^2)}{2}du^2 = \cancel{\frac{f(x^2)}{2\, dx}}$$

$$= \frac{d}{dx}\int_0^{x^2}\frac{f(\alpha)}{2}d\alpha = \frac{x}{2}f(x^2)$$

1000.9.15

① $$\int x^2\sqrt{1-x^2}\,dx$$

$$\overset{令x=\cos t}{=} \int \cos^2 t\cdot \sin t\, d\cos t = -\int(\cos t\cdot\sin t)^2 dt$$

$$= -\frac{1}{8}\int \sin^2 2t\, d2t \overset{u=2t}{=} -\frac{1}{16}\int\frac{1-\cos 4t}{2}d4t$$

$$\overset{u=4t}{=} -\frac{1}{32}\left[\int du - \int\cos u\, du\right]$$

$$= \frac{\sin 4t - 4t}{32} + C = \frac{\sqrt{1-x^2}}{8}(2x^3-x) - \frac{\arccos x}{8} + C$$

② $$\int\frac{dx}{(2x^2+1)\sqrt{x^2+1}}$$

$$\overset{x=\tan t}{=} \int\frac{\sec^2 t\, dt}{(\tan^2 t+\sec^2 t)\sec t} = \int\frac{\sec t\, dt}{2\sec^2 t-1}$$

$$= \int\frac{d\sin t}{1+\sin^2 t} \overset{\sin t=u}{=} \int\frac{du}{1+u^2} = \arctan u + C$$

$$= \arctan\frac{x}{\sqrt{1+x^2}} + C.$$

③ $\int \frac{\arcsin\sqrt{x}}{\sqrt{x}}dx$

$\overset{\sqrt{x}=t}{=} \int \frac{2t\arcsin t}{t}dt = 2\int \arcsin t\,dt$

$= 2t\arcsin t - 2\int \frac{t\,dt}{\sqrt{1-t^2}} = 2t\arcsin t - \int \frac{dt^2}{\sqrt{1-t^2}}$

$= 2t\cdot\arcsin t + \sqrt{1-t^2} = 2\sqrt{x}\cdot\arcsin\sqrt{x} - 2\sqrt{1-x} + C$

④ $\int \arccos x\,dx$

$= x\arccos x + \int \frac{x}{\sqrt{1-x^2}}dx \overset{u=x^2}{=} x\arccos x + \frac{1}{2}\int \frac{du}{\sqrt{1-u^2}}$

$= x\arccos x + \sqrt{(1-x^2)} + c$

⑤ $\int \frac{\arctan e^x}{e^x}dx \overset{\substack{e^x=t\\x=\ln t}}{=} \int \frac{\arctan t}{t}\frac{dt}{t}$

$= -\int \arctan t\, d\left(\frac{1}{t}\right) = \frac{-\arctan t}{t} + \int \frac{1}{t}\frac{1}{1+t^2}dt$

$\overset{t=\tan u}{=} \frac{\arctan t}{-t} + \int \frac{\sec^2 u\,du}{\tan u\cdot\sec^2 u} = \frac{\arctan t}{-t} + \int \frac{\cos u}{\sin u}du$

$= \frac{-\arctan e^x}{e^x} - \ln\left|\frac{1}{\sqrt{1+e^{2x}}}\right| + C$

$= \frac{-\arctan e^x}{e^x} + \frac{1}{2}\ln\left(\frac{1+e^{2x}}{e^{2x}}\right) + C$

⑥ $\int \frac{dx}{(x^2+1)(x^2+x+1)}$

$= \int \frac{\overset{-1}{A}x+\overset{0}{B}}{x^2+1}dx + \int \frac{\overset{1}{C}x+\overset{1}{D}}{x^2+x+1}dx$

$2d(x^2+x+1) = (2x+1)dx$

$= -\frac{1}{2}\int \frac{dx^2+1}{x^2+1} + \frac{1}{2}\int \frac{d\,\cancel{x^2+x+1}}{x^2+x+1}\cancel{dx} - \frac{1}{2}\int \frac{dx}{x^2+x+1}$

$= -\frac{1}{2}\ln(x^2+1) + \cancel{\int 1 - \frac{x^2}{x^2+x+1}dx}$

$\ln(x^2+x+1) - \frac{1}{2}\int \frac{d(x+\frac{1}{2})}{(x+\frac{1}{2})^2+\frac{3}{4}}$

$= -\frac{1}{2}\ln(x^2+1) + \ln(x^2+x+1) - \frac{1}{\sqrt{3}}\arctan\frac{2(x+\frac{1}{2})}{\sqrt{3}} + C$

⑦ $\int \frac{dx}{x^3+4x^2+5x+2}$

$= \int \frac{dx}{(x+1)^2(x+2)} = \int \frac{\overset{①}{A}}{x+1}dx + \int \frac{\overset{⊖①}{B}}{(x+1)^2}dx + \int \frac{\overset{(+1)}{C}}{x+2}dx$

$= \ln(x+1) + \ln(x+2) \overset{-\frac{1}{x+1}}{\wedge} + C = \ln\left|\frac{x+1}{x+2}\right| + C. - \frac{1}{x+1}$

⑧ $\int_0^{\frac{\pi}{2}} e^x\cos x\,dx$

e^x	e^x	e^x
$\cos x$	$\sin x$	$-\cos x$

$= \cancel{e^x\sin x - e^x\cos x + \int e^x}$

$\cancel{-\int e^x\cos x\,dx}$

$\Rightarrow \cancel{原 = \frac{1}{2}e^x\sin x\Big|_0^{\frac{\pi}{2}} = \frac{1}{2}e^{\frac{\pi}{2}}}$

$\Rightarrow 原 = e^x\sin x\Big|_0^{\frac{\pi}{2}} + e^x\cos x\Big|_0^{\frac{\pi}{2}} - \int e^x\cos x\,dx$

$\Rightarrow 原 = \frac{1}{2}\left[e^x\sin x\Big|_0^{\frac{\pi}{2}} + e^x\cos x\Big|_0^{\frac{\pi}{2}}\right] = \frac{1}{2}\left(e^{\frac{\pi}{2}} - 1\right)$

⑨ $\int_0^1 \frac{\ln(1+x)}{(2-x)^2}dx$

$\ln(1+x)$	$\frac{1}{x+1}$
$(2-x)^{-2}$	$(2-x)^{-1}$

$= \frac{\ln(1+x)}{2-x}\Big|_0^1 - \int_0^1 \frac{1}{x+1}\cdot\frac{1}{2-x}dx$

$= \ln 2 - 0 - \frac{1}{3}\int_0^1 \frac{1}{x+1} - \frac{1}{x-2}dx$

$= \ln 2 - \frac{1}{3}\ln\left|\frac{x+1}{x-2}\right|\Big|_0^1 = \frac{1}{3}\ln 2$

⑩ $\int_0^1 \ln(x+\sqrt{x^2+3})dx$

$\ne \cancel{\frac{1}{\sqrt{x^2+3}}\Big|_0^1 = \frac{1}{2}-\frac{1}{\sqrt{3}}}$

$= x\ln(x+\sqrt{x^2+3})\Big|_0^1 - \int_0^1 x\, d\ln(x+\sqrt{x^2+3})$

$= \ln 3 - \int_0^1 \frac{x}{\sqrt{x^2+3}}dx$

$= \ln 3 - \frac{1}{2}\int_0^1 \frac{d(u+3)}{\sqrt{u+3}} = \ln 3 - (u+3)^{\frac{1}{2}}\Big|_0^1 = \ln 3 - 2 + \sqrt{3}$

⑪ $\int_0^{+\infty} \frac{xe^{-3x}}{(1+e^{-3x})^2}dx$

$e^{-3x}=t$, $x=\frac{\ln t}{-3}$

$= \int \frac{-\frac{\ln t}{3}\, t}{(1+t)^2}\left(-\frac{1}{3}\frac{1}{t}\right)dt = -\frac{1}{9}\int_0^1 \frac{\ln t\, dt}{(1+t)^2}$

$= \frac{1}{9}\int_0^1 \ln t\, d\frac{1}{1+t} = \frac{1}{9}\frac{\ln t}{1+t}\Big|_0^1 - \frac{1}{9}\int_0^1 \frac{1}{1+t}\frac{1}{t}dt$

$= \frac{1}{9}\left(\frac{\ln t}{1+t} - \ln\frac{t}{t+1}\right)\Big|_0^1 = \frac{1}{9}\ln 2$

⑫ $\int_0^1 x\ln(1-x)dx$

$\ln(1-x)$	$\frac{-1}{1-x}$
x	$\frac{x^2}{2}$

$\cancel{= \frac{1}{2}\int_0^1 \ln(1-x)\, dx^2}$

$\cancel{= \frac{1}{2}\ln(1-x)x^2\Big|_0^1 + \frac{1}{2}\int_0^1 \frac{+x^2}{1-x}dx}$

$\cancel{= \frac{1}{2}\ln(1-x)x^2\Big|_0^1 - \frac{1}{2}\int_0^1 x^2\, d\ln(1-x)}$

$= \frac{1}{2}\ln(1-x)x^2\Big|_0^1 - \cancel{\frac{1}{2}\ln(1-x)x^2} + \int_0^1 x\ln(1-x)dx$

$= \int_0^1 (1-x)\ln(1-(1-x))dx$

$= \int_0^1 (1-x)\ln x\, dx = \int_0^1 \ln x\, dx - \int_0^1 x\ln x\, dx.$

$= \cancel{(x-1)\ln x\Big|_0^1 - \frac{x^2}{2}}(x\ln x - x)\Big|_0^1 - \left(\frac{x^2}{2}\ln x - \frac{x^2}{4}\right)\Big|_0^1$

$\lim_{x\to 0} x\ln x = \lim_{x\to 0}\frac{\ln x}{\frac{1}{x}} = \lim_{x\to 0}\frac{\frac{1}{x}}{-\frac{1}{x^2}} = 0.$

$\lim_{x\to 0} x^2\ln x = x\cdot x\ln x = 0.$

原 $= (-1-0) - \left(-\frac{1}{4}-0\right) = -\frac{3}{4}$

⑬ $\int_0^\pi \frac{x\sin x}{1+\sin^2 x}dx$

$\cancel{\sin x = t,\ x=\arcsin t\ = \int \frac{\arcsin t\cdot t}{1+t^2}\frac{dt}{\sqrt{1-t^2}}}$

$= \int_0^\pi \frac{(\pi-x)\sin x}{1+\sin^2 x}dx$

$= \frac{1}{2}\int_0^\pi \frac{\pi\sin x \cancel{- x\sin x + x\sin x}}{1+\sin^2 x}dx$

$= -\frac{\pi}{2}\int_0^\pi \frac{d\cos x}{2-\cos^2 x}$, $\cos x = t$

$= +\frac{\pi}{2}\int_{-1}^{1} \frac{dt}{(\sqrt{2})^2 - t^2}$

$= \frac{\pi}{2}\frac{1}{2\sqrt{2}}\ln\left|\frac{t+\sqrt{2}}{t-\sqrt{2}}\right|\Big|_{-1}^{1}$

$= \frac{\sqrt{2}\pi}{2}\ln\frac{(\sqrt{2}+1)}{(1-\sqrt{2})}\frac{(1+\sqrt{2})}{(1+\sqrt{2})-1}$

⑭ $\int_0^1 \arctan\sqrt{x}\, dx$

$\sqrt{x}=t$

$= \int_0^{1} \arctan t \; d t^2$

$= \arctan t \cdot t^2 \Big|_0^{1} - \int_0^{1} \frac{t^2}{1+t^2} dt$

$= \frac{\pi}{4} - \left[\int_0^1 1\, dt - \int_0^1 \frac{1}{1+t^2} dt \right]$

$= \frac{\pi}{4} - 1 + \arctan t \Big|_0^1 = \frac{\pi}{4} - 1 + \frac{\pi}{4} - 0 = \frac{\pi}{2} - 1$

⑮ $\int \frac{dx}{2+\cos x}$

$\tan\frac{x}{2} = t$

$\sin x = \frac{2t}{1+t^2}$

$\cos x = \frac{1-t^2}{1+t^2}$

$\tan x = \frac{2t}{1-t^2}$

~~$\cos x = t$~~

~~$= \int \frac{-1}{2+t} \frac{1}{\sqrt{1-t^2}} dt$~~

令 $t = \tan\frac{x}{2}$

$\int \frac{dx}{2+\cos x} = \int \frac{1}{2+\frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$

$= \int \frac{2}{3+t^2} dt = \frac{2}{3} \int \frac{1}{1+\frac{t^2}{3}} dt$

$= \frac{1}{\sqrt{3}} \arctan \frac{t}{\sqrt{3}} + C$ ~~$\frac{1}{\sqrt{3}}$~~ , $t = \tan\frac{x}{2}$

⑯ $\int_0^1 \frac{x}{\sqrt{1-x^2}} \arcsin x\, dx$

$= \int_0^1 x \arcsin x \; d \arcsin x$

$x=\sin t$

$= \int_0^{\frac{\pi}{2}} \sin t \cdot t\, dt = -t\cos t \Big|_0^{\frac{\pi}{2}} + \sin t \Big|_0^{\frac{\pi}{2}}$

$= 0+1 = 1$

⑰ $\int \frac{dx}{x+\sqrt{x+2}}$

$\sqrt{x+2} = t$

$x = t^2 - 2$

$= \int \frac{2t\,dt}{t^2-2+t} = \int \frac{2t\,dt}{(t-1)(t+2)}$

$= \int \frac{\frac{2}{3}}{t-1} dt + \int \frac{\frac{4}{3}}{t+2} dt$

$= \frac{2}{3} \ln(t-1) + \frac{4}{3} \ln(t+2) + C$

⑱ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{2^{x-1}}{2^x+1} \cos^4 2x\, dx$

~~$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\cos^4 2x \; d(2^x+1)}{2(2^x+1)\ln 2}$~~

$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{2^{-x-1}}{2^{-x}+1} \cos^4 2x\, dx$

$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{2^{-1}}{1+2^{x}} \cos^4 2x\, dx$

$= \frac{1}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\cos^4 2x}{1+2^x} dx$ ①

原 $= \frac{1}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{2^x}{1+2^x} \cos^4 2x\, dx$ ②

由①② 原 $= \frac{1}{4} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1+2^x}{1+2^x} \cos^4 2x\, dx = \frac{1}{4} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^4 2x\, dx$

$= \frac{1}{4} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{3}{8} + \frac{\cos 8x}{4} dx = \frac{3}{64}\pi$

⑲ $\int \frac{x^2}{\sqrt{4-x^2}}\,dx$

$= \int \frac{-x}{\sqrt{4-x^2}}\,x\,dx = \int x\,d\sqrt{4-x^2}$

$= x\sqrt{4-x^2} - \int \sqrt{4-x^2}\,dx = x\sqrt{4-x^2} - \frac{4}{2}\arcsin\frac{x}{2} - \frac{x}{2}\sqrt{4-x^2}$

$= \frac{1}{2}x\sqrt{4-x^2} - 2\arcsin\frac{x}{2} + C$

⑳ $\int_0^2 [(x-1)^3+2x]\sqrt{1-\cos 2\pi x}\,dx$

$= \int_0^2 (1-x)^3 + 2(2-x)\sqrt{1-\cos 2\pi(2-x)}\,dx$

$= \int_0^2 (1-x)^3 + 2(2-x)\sqrt{1-\cos 2\pi x}\,dx$

$= \frac{4}{2}\int_0^2 \sqrt{1-\cos 2\pi x}\,dx = 2\int_0^2 \sqrt{1-\cos 2\pi x}\,dx.$

~~$\cos 2\pi x = t$~~ $\tan \pi x = t$

$\cos 2\pi x = \frac{1-t^2}{1+t^2}$

$x = \frac{1}{\pi}\arctan t.$

$2\int \sqrt{1-\frac{1-t^2}{1+t^2}}\cdot\frac{1}{\pi}\cdot\frac{1}{1+t^2}\,dt$ ($1+t^2-1+t^2$)

$= \frac{2}{\pi}\int \frac{\sqrt{2}\,t\,dt}{(1+t^2)^{\frac{3}{2}}}$

$= \int \frac{1}{\pi}\frac{\sqrt{2}\,d(t^2+1)}{(1+t^2)^{\frac{3}{2}}} \xrightarrow{u=t^2+1} \int \frac{\sqrt{2}}{\pi}\frac{du}{u^{\frac{3}{2}}}$

$= \frac{\sqrt{2}}{\pi}(-2)\,u^{-\frac{1}{2}}\Big|_{x=0}^{x=2} = \frac{-2\sqrt{2}}{\pi}(\tan^2\pi x + 1)\Big|_0^2 = 0$

$= 2\int_0^2 \sqrt{\frac{1-\cos 2\pi x}{2}}\,\sqrt{2}\,dx$

$= 2\sqrt{2}\int_0^2 |\sin \pi x|\,dx = 4\sqrt{2}\int_0^1 \sin \pi x\,dx = \frac{8\sqrt{2}}{\pi}$

㉑ $\int_0^{+\infty} \frac{1}{(x^2+1)(1+x^5)}\,dx$

$x=\tan t$

$= \int_0^{\frac{\pi}{2}} \frac{\sec^2 t\,dt}{(\tan^2 t+1)(1+\tan^5 t)}$

$= \int_0^{\frac{\pi}{2}} \frac{\cos^5 t\,dt}{(\sin^2 t+\cos^2 t)(\sin^5 t+\cos^5 t)}$

$= \int_0^{\frac{\pi}{2}} \frac{\cos^5 t\,dt}{\sin^5 t+\cos^5 t}$

$= \int_0^{\frac{\pi}{2}} \frac{\cos^5(\frac{\pi}{2}-t)\,dt}{\sin^5(\frac{\pi}{2}-t)+\cos^5(\frac{\pi}{2}-t)} = \int_0^{\frac{\pi}{2}} \frac{\sin^5 t\,dt}{\cos^5 t+\sin^5 t}$

$\Rightarrow$ 原 $= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{\cos^5 t+\sin^5 t}{\cos^5 t+\sin^5 t}\,dt = \frac{\pi}{4}$

㉒ $\int \frac{4x}{(2x+1)^3}\,dx$

$4x$	4	0
$\frac{1}{(2x+1)^3}$	$\frac{(2x+1)^{-2}}{-2\times 2}$	$\frac{(2x+1)^{-1}}{(-2)(-1)\times 4}$

$= -\frac{2x(2x+1)^{-2}}{2} - \frac{2(2x+1)^{-1}}{4} + C$

$= -x(2x+1)^{-2} - \frac{1}{2}(2x+1)^{-1} + C$

㉓ $\int \frac{x^3}{(x-1)^{100}}dx = \int \frac{[(x-1)+1]^3}{(x-1)^{100}}dx$

$= \int \frac{(x-1)^3+3(x-1)^2+3(x-1)+1}{(x-1)^{100}}dx$

$= \int \frac{d(x-1)}{(x-1)^{97}} + 3\int \frac{d(x-1)}{(x-1)^{98}} + 3\int \frac{d(x-1)}{(x-1)^{99}} + \int \frac{d(x-1)}{(x-1)^{100}}$

$= \frac{(x-1)^{-96}}{-96} + 3\frac{(x-1)^{-97}}{-97} + 3\frac{(x-1)^{-98}}{-98} + \frac{(x-1)^{-99}}{-99} + C$

㉔ $\int_1^{+\infty} \frac{dx}{x^2(x+1)}$

$= \int_1^{+\infty} \frac{d\frac{1}{x}}{-(x+1)} = -\frac{1}{x(x+1)}\Big|_1^{+\infty} + \int_1^{+\infty} \frac{1}{x}\, d\frac{1}{x+1}$

$= \frac{1}{2} - \int_1^{+\infty} \frac{1}{x(x+1)^2}dx$

$\underline{t=x+1}$ $\frac{1}{2} - \int_2^{+\infty} \frac{dt}{t^2(t-1)}$

$= \int_1^{+\infty} \left(\frac{1}{x^2} - \frac{1}{x} + \frac{1}{x+1}\right)dx.$

$= 1 - \ln 2$

㉕ $\int \frac{\cos\theta}{\sin\theta - 2\cos\theta}d\theta$

$= \int \frac{1}{\tan\theta - 2}d\theta$

$\underline{\tan\theta = t}$ $\int \frac{1}{t-2}\frac{1}{1+t^2}dt.$

$= \int \frac{-\frac{t}{5}-\frac{2}{5}}{1+t^2} + \frac{\frac{1}{5}}{t-2}dt = -\frac{1}{5}\int \frac{t+2}{1+t^2} - \frac{1}{t-2}dt$

$= \frac{1}{5}\left(-2\arctan t - \frac{1}{2}\ln[1+t^2] + \ln|t-2|\right)$

$= \frac{1}{5}\left(\ln|\sin\theta - 2\cos\theta| - 2\theta\right) + C.$

㉖ $\int \frac{2x+1}{x^2}e^{-2x}dx$

$= 2\int \frac{e^{-2x}}{x}dx + \int \frac{e^{-2x}}{x^2}dx$

$= \frac{2}{-2}\int \frac{1}{x}\, de^{-2x} + \int \frac{e^{-2x}}{x^2}dx$

$= -\frac{1}{x}e^{-2x} - \int \frac{e^{-2x}}{x^2}dx + \int \frac{e^{-2x}}{x^2}dx = -\frac{1}{x}e^{-2x} + C$

㉗ $\int \frac{x+\sin x}{1+\cos x}dx$

$= \int \frac{x}{1+\cos x}dx + \int \frac{\sin x}{1+\cos x}dx$

$= \int \frac{x}{1+\cos x}dx + \frac{x\sin x}{1+\cos x} - \int x\, d\frac{\sin x}{1+\cos x}$

$= \int \frac{x}{1+\cos x}dx + \frac{x\sin x}{1+\cos x} - \int \frac{x}{1+\cos x}dx$

$= \frac{x\sin x}{1+\cos x} + C$

1000.9.16.

$f(x)=\frac{e^x+e^{-x}}{2}$ 求 $\int [\frac{f'(x)}{f(x)}+\frac{f(x)}{f'(x)}]dx$

$f(x)=\frac{e^x+e^{-x}}{2}$ $\quad \frac{f'(x)}{f(x)}=\frac{e^x-e^{-x}}{e^x+e^{-x}}=\frac{e^{2x}-1}{e^{2x}+1}$

$f'(x)=\frac{e^x-e^{-x}}{2}$

$\frac{f'(x)}{f(x)}+\frac{f(x)}{f'(x)}=\frac{e^{2x}-1}{e^{2x}+1}+\frac{e^{2x}+1}{e^{2x}-1}=2\frac{e^{4x}+1}{e^{4x}-1}$

$=2\frac{e^{2x}+e^{-2x}}{e^{2x}-e^{-2x}}$ $\quad \cancel{=2f}$

$\int G(x)dx \overset{e^{2x}=t}{\underset{x=\frac{\ln t}{2}}{=}} 2\int \frac{t^2+1}{t^2-1}\frac{1}{2}\frac{dt}{t}$

$=\cancel{\int \frac{t^2+1}{t(t+1)(t-1)}dt}=\int \frac{t^2+1}{t(t^2-1)}dt$

$\underline{\underline{t=\sec u}} \int \frac{(\sec^2 u+1)\sec u \tan u\, du}{\sec u \tan^2 u}$

$=\int \frac{(\sec^2 u+1)du}{\tan u}=\int \frac{d\tan u}{\tan u}+\int \cot u\, du$

$=\ln|\tan u|+\ln|\sin u|+C$

$=\ln|\frac{t^2-1}{t}|+C=\ln|e^{2x}-e^{-2x}|+C$ ✓

$f(x)=f''(x)$!

原$=\int [\frac{f'}{f}+\frac{f}{f'}]dx=\int d[\ln f+\ln|f'|]=\ln[ff']+C$

1000.9.17

设 $f(x)$ 一个原函数 $F(x)=\ln^2(x+\sqrt{1+x^2})$ 求 $\int xf'(x)dx$

$\int xf'(x)dx=\int x\,df(x)=xf(x)-\int f(x)dx$

$\Rightarrow f(x)=\frac{2\ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}}$

原 $=\frac{2x\ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}}-\ln^2(x+\sqrt{1+x^2})+C$

1000.9.18

$I=\int_0^1 \frac{f(x)}{\sqrt{x}}dx$, 其中 $f(x)=\int_1^{\sqrt{x}} e^{-t^2}dt$

[solution]. $f'(x)=e^{-x}$.

$I=\int_0^1 \frac{f(x)}{\sqrt{x}}dx=+2\int_0^1 f(x)\,dx^{\frac{1}{2}}$ $\quad f'(x)=\frac{e^{-x}}{2\sqrt{x}}$

$=2\sqrt{x}f(x)\Big|_0^1-2\int_0^1 \frac{\sqrt{x}e^{-x}}{2\sqrt{x}}dx=0-\int_0^1 e^{-x}dx=e^{-1}-1$

$\cancel{\sqrt{x}=t}$

$\cancel{=2f(1)-2\int_0^1 2t e^{-t^2}dt}$

$\cancel{=-4\int_0^1 t^2 e^{-t^2}dt}$

$\cancel{=+2\int_0^1 t\cdot de^{-t^2}}$

$\cancel{=2te^{-t^2}\Big|_0^1-2\int_0^1 e^{-t^2}dt}$

$\cancel{t \quad 2t \quad e^{-t^2} \quad ?}$ $\quad t^2$

1000.9.19

$$\underline{\int_0^{\pi} e^{-x}\sin nx\,dx} \qquad \begin{array}{c|c|c} \sin nx & n\cos nx & -n^2\sin nx \\ e^{-x} & -e^{-x} & e^{-x} \end{array}$$

$$= -e^{-x}\sin nx\Big|_0^{\pi} - ne^{-x}\cos nx\Big|_0^{\pi} + n^2\underline{\int_0^{\pi} \sin nx\, e^{-x}\,dx}$$

$$\Rightarrow \int_0^{\pi} e^{-x}\sin nx\,dx = \frac{e^{-x}\sin nx + ne^{-x}\cos nx}{n^2-1}\Bigg|_0^{\pi}$$

$$= \frac{1}{n^2-1}\left[ne^{-x}\cos nx - ne^{-x}\right]$$

$$= \frac{n}{n^2-1}\left(e^{-\pi}(-1)^{n+1} - 1\right)$$

1000.9.20

$$g(x) = \int_0^x f(t)\,dt, \quad f(x) = \begin{cases} \frac{1}{2}(x^2+1) & 0 \le x < 1 \\ \frac{1}{3}(x-1) & 1 \le x \le 2 \end{cases}$$

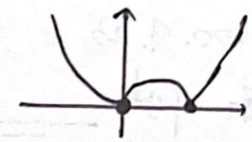
求 $g(x)$ 并判定 $[0,2]$ 连续与可导性.

$$g(x) = \begin{cases} \frac{1}{2}\left[\frac{x^3}{3} + x\right], & 0 \le x < 1 \\ \frac{2}{3} + \frac{1}{3}\left[\frac{x^2}{2} - x\right], & 1 \le x \le 2 \end{cases} \qquad \int_0^1 f(t)\,dt = g(1) = \frac{2}{3}$$

连续✓ $\lim\limits_{x\to 1^-} g'(x) = \frac{1}{2}[x^2+1]\Big|_{x=1} = 1$

$\lim\limits_{x\to 1^+} g'(x) = \frac{1}{3}(1-1) = 0.$ 不可导.

1000.9.21



$$\int_0^2 |x-x^2|\,dx = \int_0^2 |x^2-x|\,dx$$

$$= \int_1^2 (x^2-x)\,dx - \int_0^1 (x^2-x)\,dx$$

$$= \left(\frac{x^3}{3} - \frac{x^2}{2}\right)\Big|_1^2 - \left(\frac{x^3}{3} - \frac{x^2}{2}\right)\Big|_0^1 = \left(\frac{2}{3} + \frac{1}{6}\right) + \frac{1}{6} = 1$$

1000.9.22

已知 $f(x)$ 为连续函数, $\int_0^x t f(x-t)\,dt = 1 - \cos x$.
求 $\int_0^{\frac{\pi}{2}} f(x)\,dx$

① $\int_0^x t f(x-t)\,dt = \int_0^x (x-t) f(x-x+t)\,dt = \int_0^x (x-t) f(t)\,dt$

$= x\int_0^x f(t)\,dt - \int_0^x t f(t)\,dt = 1 - \cos x$

$x = \frac{\pi}{2}$ $\Rightarrow \cancel{\frac{\pi}{2}\int_0^{\frac{\pi}{2}} f(x)\,dx - \int_0^{\frac{\pi}{2}} x f(x)\,dx = 1 - \cos\frac{\pi}{2} = 1}$

$\Rightarrow \cancel{\frac{\pi}{2}\int_0^{\frac{\pi}{2}} f(x)\,dx = 1 + \int_0^{\frac{\pi}{2}} x f(x)\,dx}$

② $x f(x) + \int_0^x f(t)\,dt - x f(x) = \sin x$ ← 求导

$x = \frac{\pi}{2}$ $\int_0^{\frac{\pi}{2}} f(x)\,dx = 1.$

1000.9.23.

求 $\int_{e^{\frac14}}^{e^{\frac12}} \frac{dx}{x\sqrt{\ln x\,(1-\ln x)}}$

$$= \int_{e^{\frac14}}^{e^{\frac12}} \frac{d\ln x}{\sqrt{\ln x\,(1-\ln x)}} \overset{\ln x = t}{\underset{x=e^t}{=}} \int_{\frac14}^{\frac12} \frac{dt}{\sqrt{t(1-t)}}$$

$$= \int_{\frac14}^{\frac12} \frac{dt}{\sqrt{(\frac12)^2-(t-\frac12)^2}} = \cancel{\ln\left(t-\tfrac12+\sqrt{(\tfrac12)^2-(t-\tfrac12)^2}\right)\Big|_{\frac14}^{\frac12}}$$

$$= \cancel{\ln 2}\ \arcsin\frac{t-\frac12}{\frac12}\Big|_{\frac14}^{\frac12} = (0+\frac{\pi}{6}) = \frac{\pi}{6}$$

1000.9.24

设 $f(x)=\begin{cases}\frac{1}{1+x}, & x\geq 0\\ \frac{1}{1+e^x}, & x<0.\end{cases}$ 求 $\int_0^2 f(x-1)\,dx$.

$$\int_0^2 f(x-1)\,dx \overset{x-1=t}{\underset{x=1+t}{=}} \int_{-1}^{1} f(t)\,dt$$

$$= \int_{-1}^{1} f(x)\,dx = \int_{-1}^{0}\frac{1}{1+e^x}dx + \int_0^1 \frac{1}{1+x}dx$$

$$= \int_{-1}^{0}\frac{de^x}{e^x(1+e^x)} + \ln|1+x|\Big|_0^1$$

$$= \ln 2 + \int_{e^{-1}}^{1}\frac{dt}{t(1+t)} = \ln 2 + \int_{e^{-1}}^{1}\frac{dt+\frac12}{(t+\frac12)^2-\frac14} = \ln 2 + \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|_{x=e^{-1}+\frac12}^{x=1+\frac12}$$

$a=\frac12$

$= \ln(1+e)$?

1000.9.25

$$\int_{-2}^{2} e^{|x|}(x+x^2)\,dx$$

$$= \int_{-2}^{2} e^{|x|}(-x+x^2)\,dx$$

x^2	$2x$	2	0
e^x	e^x	e^x	e^x

$$\rightarrow 原 = \frac12\int_{-2}^{2} e^{|x|}x^2\,dx = \int_0^2 e^x x^2\,dx$$

$$= x^2e^x - 2xe^x + 2e^x\Big|_0^2$$

1000.9.26

求 $I=\int_0^{\ln 2}\frac{xe^x}{e^x+1}dx + \int_{\ln 2}^{\ln 3}\frac{xe^x}{e^x-1}dx$

$$\cancel{= \int_0^{\ln 3}\frac{xe^x(e^x-1)}{e^{2x}-1} + \frac{xe^x(e^x+1)}{e^{2x}-1}dx}$$

$2x=t$

$$\cancel{= \int_0^{\ln 3}\frac{xe^x\cdot 2e^x}{e^{2x}-1}dx} = \frac12\int_0^{\ln 3}\frac{2xe^{2x}}{e^{2x}-1}dx$$

$$\cancel{= \frac12\int_0^{2\ln 3}\frac{te^t\,dt}{e^t-1}} \overset{e^t=u}{\underset{t=\ln u}{=}} \frac12\int_1^9 \frac{\ln u\cdot u}{(u-1)u}du$$

$e^{\ln 9}=u$

$$\cancel{= \frac12\int_1^9\frac{\ln u}{u-1}du \overset{m=u-1}{=} \frac12\int_0^8\frac{\ln(m+1)}{m}dm}$$

$= \cancel{\frac12}$

$$\overset{e^x=u}{I}=\int_1^2 \frac{\ln u}{u+1}du+\int_2^3\frac{\ln u}{u-1}du=I_1+I_2.$$

对I_2: $u-1=t$. $u=t+1$

$$I_2=\int_1^2\frac{\ln(t+1)}{t}dt=\ln(t+1)\ln t\Big|_1^2-\int_1^2\frac{\ln t}{t+1}dt$$

$$=\ln2\ln3-I_1$$

$$\therefore I=I_1+I_2=\ln2\ln3.$$

1000. 9t.1

$$\begin{cases}a_n=\int_0^1 x^n\sqrt{1-x^2}\,dx,\\ b_n=\int_0^{\frac{\pi}{2}}\sin^n t\,dt\end{cases}\qquad \lim_{n\to\infty}\frac{na_n}{b_n}=?$$

[solution]

$$a_n\xlongequal{x=\sin t}\int_0^{\frac{\pi}{2}}\sin^n t\cdot\cos t\,d\sin t=\int_0^{\frac{\pi}{2}}\sin^n t\cdot(1-\sin^2 t)dt$$

$$=\int_0^{\frac{\pi}{2}}\sin^n t-\sin^{n+2}t\,dt.$$

$$\lim_{n\to\infty}\frac{na_n}{b_n}=\lim_{n\to\infty}\frac{n(\frac{-n+1}{n+2}+1)(\cancel{\frac{n-1}{n}\frac{n-3}{n-2}}\cdots)}{(\cancel{\frac{n-1}{n}\frac{n-3}{n-2}}\cdots)}=\lim_{n\to\infty}\frac{n(\overset{n+2-n+1}{1})}{n+2}=1$$

1000. 9t.2

$f(x)$连续导数, $x>0$时, 满足$f(\ln x)=\frac{1}{x^2}$,

$$\int_0^1 xf(x)dx=?$$

$$e^t=x\leftarrow t=\ln x\qquad f(t)=\frac{1}{e^{2t}}$$

$$\int_0^1 x\,df(x)=xf(x)\Big|_0^1-\int_0^1 f(x)dx$$

$$=\frac{x}{e^{2x}}\Big|_0^1-\int_0^1 e^{-2t}dt$$

$$=e^{-2}+\frac{e^{-2t}}{2}\Big|_0^1=e^{-2}+\frac{e^{-2}}{2}-\frac{1}{2}=\frac{3e^{-2}-1}{2}$$

1000. 9t.3

可导函数$y=f(x)$在$[0,+\infty)$值域$[0,+\infty)$, $f(0)=0$,

$f'(x)>0$. $x=\varphi(y)$是$y=f(x)$反函数. $a\cdot b>0$.

$$I=\int_0^a f(x)dx+\int_0^b\varphi(y)dy,\quad a<\varphi(b)\text{时}$$

$$\cancel{=\int_0^a f(x)dx-\int_{\substack{x=0\\y=0}}^{\substack{x=\varphi(b)\\y=b}}\frac{\varphi(y)}{f'(x)}dx}\qquad \cancel{a<\varphi(b).}$$

~~找到 $f(x)$ 与 $\frac{\varphi(y)}{f'(x)}$　$f(x)\cdot f'(x)=\varphi(y)$~~

$$I-ab=\int_0^a f(x)dx+\int_0^{\varphi(b)}\varphi[f(x)]\,d[f(x)]-ab$$

$$=\int_0^a f(x)dx+\int_0^{\varphi(b)}x\,d[f(x)]-ab$$

$$=\int_0^a f(x)dx+xf(x)\Big|_0^{\varphi(b)}-\int_0^{\varphi(b)}f(x)dx-ab$$

$$=\int_{\varphi(b)}^a f(x)dx+\varphi(b)\cdot b-0-ab.$$

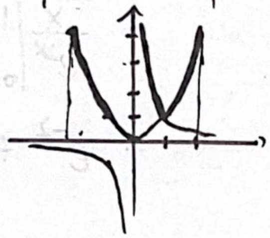
$$=\int_{\varphi(b)}^a f(x)dx-\int_{\varphi(b)}^a b\,dx=\int_{\varphi(b)}^a[f(x)-b]dx$$

$$a<x<\varphi(b)\Rightarrow f(x)<f[\varphi(b)]=b\Rightarrow I-ab>0.$$

1000.9t.4

$$\int_{-2}^{2} \max\left\{x^2, \frac{1}{\sqrt[3]{x^2}}\right\}dx = \int_{-2}^{2} \max\left\{x^2, x^{-\frac{3}{2}}\right\} dx$$

$$= \int_{0}^{2} x^2 dx + \int_{1}^{2} x^2 dx + \int_{0}^{1} x^{-\frac{3}{2}} dx$$



$$= \frac{x^3}{3}\Big|_0^2 + \frac{x^3}{3}\Big|_1^2 + \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}}\Big|_0^1$$

$$= \frac{9}{3} + \frac{9}{3} - \frac{1}{3} - (-2-0) = \frac{23}{3}$$

1000.9t.5

$$\begin{cases} x = a(t-\sin t) \\ y = a(1-\cos t) \end{cases} \quad (0 \le t \le \pi) \qquad \int_0^{\pi a} y^3 dx = ?$$

[solution]. $\int_0^{\pi} a^3(1-\cos t)^3 \cdot d\, a(t-\sin t)$

$$= a^4 \int_0^{\pi} (1-\cos t)^4 dt$$

$$= a^4 \int_0^{\pi} \cos^4 t - 4\cos^3 t + 6\cos^2 t - 4\cos t + 1\, dt$$

$$= a^4\left[\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} - 0 + 6\cdot\frac{1}{2}\cdot\frac{\pi}{2} - 0 + \pi\right]$$

$$= a^4 \frac{35}{8}\pi$$

1000.9t.6

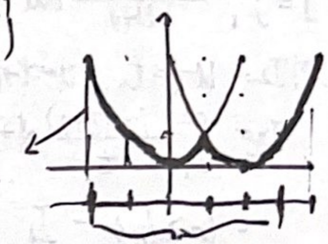
$f(x) = \min\{(x-k)^2, (x-k-2)^2\}$.

$g(k) = \int_0^1 f(x)dx$. 求 $g(k)$在 $-2<k\le 2$上最值

([分析])

$f(x) = \min\{(x-k)^2, (x-k-2)^2\}$

$f(x+k) = \min\{x^2, (x-2)^2\}$



$g(k) = \int_0^1 f(x)dx$ $\quad v(x)$

$\Rightarrow u(x) = \int_x^{x+1} v(t)dt \quad x\in(-2,1)$.

max: $\int_{-2}^{1} x^2 dx = \frac{x^3}{3}\Big|_{-2}^{1} = \frac{1}{3} + \frac{8}{3} = 3$

min $\int_{-\frac{1}{2}}^{\frac{1}{2}} x^2 dx = 2\cdot\frac{x^3}{3}\Big|_0^{\frac{1}{2}} = \frac{1}{12}$

1000.9t.7

求$\int_0^x f(t)g(x-t)dt$ 其中 $x\ge 0$时 $f(x)=x$ 且

$$g(x) = \begin{cases} \sin x, & 0 \le x < \frac{\pi}{2} \\ 0, & x \ge \frac{\pi}{2}. \end{cases}$$

[solution].

$x-(x-t)$

$$\int_0^x f(t)g(x-t)dt$$

$$= \int_0^x f(x-t)g(t)dt = \int_0^x (x-t)g(t)dt$$

$$= x\int_0^x g(t)dt - \int_0^x t g(t)dt$$

$$= \begin{cases} x - x\cos x - (-x\cos x + \sin x) = x - \sin x, & x\in[0,\frac{\pi}{2}) \\ \cancel{\sin x\Big|_{x=\frac{\pi}{2}}} = \int_0^{\frac{\pi}{2}}(x-u)\sin u\, du = x-1. & x \ge \frac{\pi}{2} \end{cases}$$

1000.题8

设 $I_n=\int_0^{\frac{\pi}{4}}\tan^n x\,dx$

证明(1). $I_n+I_{n-2}=\frac{1}{n-1}\ (n\geq 2)$ 并求 I_n.

$I_n=\int_0^{\frac{\pi}{4}}\tan^n x\,dx$

$I_{n-2}=\int_0^{\frac{\pi}{4}}\tan^{n-2} x\,dx$

$I_n+I_{n-2}=\int_0^{\frac{\pi}{4}}\tan^{n-2}x(1+\tan^2 x)\,dx$

$=\int_0^{\frac{\pi}{4}}\tan^{n-2}x\,d\tan x=\int_0^1 u^{n-2}du=\frac{1}{n-1}$

(2). 证明 $\frac{1}{2(n+1)}<I_n<\frac{1}{2(n-1)}\ (n\geq 2)$.

即证: $\frac{1}{2}(I_{n+2}+I_n)<I_n<\frac{1}{2}(I_n+I_{n-2})$

即证 $I_{n+2}<I_n<I_{n-2}$

$\because \tan^n x\in \cancel{(0,\ )}\ (0,1),\ x\in(0,\frac{\pi}{4})$

$\therefore \tan^{n+2}x<\tan^n x<\tan^{n-2}x.$

$\therefore \int_0^{\frac{\pi}{4}}\tan^{n+2}x\,dx<\int_0^{\frac{\pi}{4}}\tan^n x\,dx<\int_0^{\frac{\pi}{4}}\tan^{n-2}x\,dx$ 得证

1000.题9

$I_n=\int_{-1}^1(x^2-1)^n dx=\int_{-1}^1(x^2-1)^n dx$

~~令 $x=\sec t$~~ $\cancel{\int_{x=-1,\,t=-\frac{\pi}{2}}^{x=1,\,t=\frac{\pi}{2}}\tan^{n+2}t\cdot dt=2\int_0^{\frac{\pi}{2}}\tan^{n+2}t\,dt}$

$I_n=x(x^2-1)^n\Big|_{-1}^1-\int_{-1}^1 x\,d(x^2-1)^n$

$=-\int_{-1}^1 nx(x^2-1)^{n-1}(2x)\,dx$

$=-2n\int \underbrace{x^2}_{[(x^2-1)+1]}(x^2-1)^{n-1}dx$

$=-2n\int_{-1}^1(x^2-1)^n dx-2n\int_{-1}^1(x^2-1)^{n-1}dx$

$\Rightarrow I_n=-2nI_n-2nI_{n-1}$

$\Rightarrow I_n=\frac{-2n}{2n+1}I_{n-1}=\frac{-2n}{2n+1}\cdot\frac{-2(n-1)}{2n-1}I_{n-2}=\cdots$

$I_2=-\frac{4}{5}I_1\qquad I_1=\int_{-1}^1(x^2-1)dx=-\frac{4}{3}$

$I_n=(-1)^n\frac{2^{2n+1}(n!)^2}{(2n+1)!}$ ✓

1000.题10

$\int_{e^{-2n\pi}}^1\left|[\cos(\ln\frac{1}{x})]'\right|\ln\frac{1}{x}\,dx.$

$\ln\frac{1}{x}=t$

$\frac{1}{x}=e^t$

$x=e^{-t}$

$\cancel{=\int_{2n\pi}^0\left|[\cos t]'\right|t\,e^{-t}(-1)dt}$

$=\int_{e^{-2n\pi}}^1\ln\frac{1}{x}\,d\left|\cos(\ln\frac{1}{x})\right|$

$=\ln\frac{1}{x}\cos(\ln\frac{1}{x})\Big|_{e^{-2n\pi}}^1-\int_{e^{-2n\pi}}^1\left|\cos(\ln\frac{1}{x})\right|\cdot\frac{x}{-x^2}\,dx$

$-\frac{1}{x}$

$=-2n\pi+\int_{e^{-2n\pi}}^1\left|\cos(\ln\frac{1}{x})\right|d\frac{1}{x}=-2n\pi+\int_{2n\pi}^0\cos(\ln t)\,dt$

$\ln\frac{1}{x}=t$

原 $=\int_0^{2n\pi}\left|\frac{d(\cos t)}{dt}\cdot\frac{dt}{dx}\right| t\cdot e^{-t}dt$

$=\int_0^{2n\pi}|e^t\cdot\sin t|\ t\cdot e^{-t}dt=\int_0^{2n\pi}|\sin t|t\,dt$

$=\sum_{k=1}^{2n}\int_{(k-1)\pi}^{k\pi}(-1)^{k-1}t\sin t\,dt.$

$=\sum_{k=1}^{2n}(-1)^{k-1}(-t\cos t+\sin t)\Big|_{(k-1)\pi}^{k\pi}$

$=\sum_{k=1}^{2n}(-1)^{k-1}[-k\pi(-1)^k+(k-1)\pi(-1)^{k-1}]$

$=\sum_{k=1}^{2n}(2k-1)\pi=4n^2\pi.$

1000.9t.11

设 n 为正整数. $I_n=\int_0^{\frac{\pi}{2}}\frac{\sin 2nx}{\sin x}dx$ $\left(=\int_0^{\frac{\pi}{2}}\frac{\sin(2n\frac{x}{\pi}-n\pi)}{\cos x}dx=\int_0^{\frac{\pi}{2}}\frac{\sin(2n(\frac{\pi}{2}-x))}{\sin(\frac{\pi}{2}-x)}dx\right)$

证明 $I_n-I_{n-1}=(-1)^{n-1}\cdot\frac{2}{2n-1}$ $(n\geqslant 2)$.

$I_{n-1}=\int_0^{\frac{\pi}{2}}\frac{\sin(2n-2)x}{\sin x}dx=\int_0^{\frac{\pi}{2}}\frac{\sin(2nx-2x)}{\sin x}dx$

$=\cancel{\int_0^{\frac{\pi}{2}}\frac{\sin 2nx\cos 2x-\cos 2nx\sin 2x}{\sin x}}$

$I_n-I_{n-1}=\int_0^{\frac{\pi}{2}}\frac{\sin 2nx-\sin(2nx-2x)}{\sin x}dx$

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$\int_0^{\frac{\pi}{2}}\frac{\sin((2n-1)x+x)-\sin((2n-1)x-x)}{\sin x}dx$

$=2\int_0^{\frac{\pi}{2}}\frac{\cos((2n-1)x)\cdot\cancel{\sin x}}{\cancel{\sin x}}dx$

$=2\frac{\sin((2n-1)x)}{2n-1}\Big|_0^{\frac{\pi}{2}}=\frac{2}{2n-1}(-1)^{n-1}$

(2) $\int_0^{\frac{\pi}{2}}\frac{\sin 6x}{\sin x}dx=I_3.$

$I_2=\int_0^{\frac{\pi}{2}}\frac{\sin 4x}{\sin x}$ $\quad I_1=\int_0^{\frac{\pi}{2}}\frac{\sin 2x}{\sin x}dx$ $\quad \sin x\Big|_0^{\frac{\pi}{2}}$

$=\int_0^{\frac{\pi}{2}}2\cos x\,dx=2$

$I_2=2+(-1)^1\frac{2}{3}=\frac{4}{3}$

$I_3=\frac{4}{3}+(-1)^2\frac{2}{5}=\frac{26}{15}$

1000.9t.12

$\int_0^1\frac{x^b-x^a}{\ln x}dx$ $(a,b>0)$.

$=\cancel{\int_0^1\int_a^b}\ \int_0^1\int_a^b x^y\,dy\,dx$

$=\int_a^b\int_0^1 x^y\,dx\,dy=\int_a^b\frac{1}{y+1}dy=\ln\frac{b+1}{a+1}$

1000. 9t-13.

证明 $\int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^\lambda} dx = \int_0^{\frac{\pi}{2}} \frac{1}{1+(\cot x)^\lambda} dx = \frac{\pi}{4}$

令 $\tan x = t$.

$\int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^\lambda} dx = \int_0^{+\infty} \frac{1}{1+t^\lambda} \cdot \frac{1}{1+t^2} dt$

令 $\cot x = u$

$\int_0^{\frac{\pi}{2}} \frac{1}{1+(\cot x)^\lambda} dx = \int_{+\infty}^{\cancel{+\infty}} \cancel{\frac{1}{1+t^\lambda}} \cdot \frac{1}{1+u^2} du = \int_0^{+\infty} \frac{1}{1+u^\lambda} \frac{1}{1+u^2} du$

$I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^\lambda} + \frac{1}{1+(\cot x)^\lambda} dx$

$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{1+(\tan x)^\lambda}{1+(\tan x)^\lambda} dx = \frac{\pi}{4}$

1000. 9t-14

设 $f(x) = \begin{cases} \frac{2+x(\arcsin x)^2}{\sqrt{4-x^2}}, & -1 \le x \le 1 \\ \frac{\arctan x}{x^2}, & x > 1. \end{cases}$

求 $\int_{-1}^{+\infty} f(x) dx$

[solution].

① $\int_{-1}^{1} f(x) dx = \underbrace{\int_{-1}^{1} \frac{x(\arcsin x)^2}{\sqrt{4-x^2}} dx}_{0} + \int_{-1}^{1} \frac{2}{\sqrt{4-x^2}} dx = \int_{-1}^{1} \frac{1}{\sqrt{1-(\frac{x}{2})^2}} dx$

$= +\frac{2}{3}\pi$

② $\int_1^{+\infty} f(x) dx = \int_1^{+\infty} \frac{\arctan x}{x^2} dx$

$= -\int_1^{+\infty} \arctan x \, d\frac{1}{x} = -\frac{\arctan x}{x} \Big|_1^{+\infty} + \int_1^{+\infty} \frac{1}{x} d\arctan x$

$= 0 + \frac{\arctan 1}{1} + \int_1^{+\infty} \frac{1}{x(1+x^2)} dx$

$x = \tan t$

$= \frac{\pi}{4} + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\tan t \cdot \sec^2 t} d\tan t$

$= \frac{\pi}{4} + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{dt}{\tan t} = \frac{\pi}{4} + \ln|\sin t| \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi}{4} + \frac{1}{2}\ln 2$